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National Security Implications of Contemporary Geo-politics and Threat Intelligence

Assessment 2: Project Report

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1. Introduction

Currently, technology is increasingly integrated into many different systems from individuals to large enterprises, including government agencies. The development of a series of modern technologies has made it easier for people to manage and replace many operations that require human intervention. However, along with the development of new technologies, there is an increase in cyber-attacks that are taking place every day around the world. It not only increases in number but also increases in the complexity and sophistication of the attacks. Critical infrastructure, businesses, and even government agencies are often the targets of malware attacks and many other sophisticated attacks (Joshi et al., 2025; CyberCX, 2025). Moreover, geopolitical tensions are also one of the causes of the increase in the number of attacks, which weakens the defence capabilities of the system and data security (Almahmoud et al., 2025)

Electricity, water, education and transportation are among the sectors that are frequently targeted by attacks because the disruption of one of the above sectors will be a direct threat to national security and social life. According to statistics from the Australian Signals Authority (2024), within 2 years from 2023 to 2024, more than 11% of cyber-attacks were recorded targeting these key sectors. Many new vulnerabilities have been discovered in the context of many operational technologies (OT) and information technology (IT) in the sectors applying and integrating many new technologies in addition to improving operational efficiency. The emergence of many modern vulnerabilities has somewhat reduced the ability to adapt and respond promptly to incidents and threats from outside (Australian Signals Authority, 2024). By exploiting readily available vulnerabilities combined with advanced and sophisticated tools, attackers can easily carry out attacks that can have devastating consequences ranging from the theft of sensitive data to prolonged disruption of essential services with geopolitical motives.

Traditional information security and defence systems are outdated or no longer effective in preventing emerging attacks. Facing the development of threats, organisations need to be equipped with a solution that helps increase defence capabilities, proactively detect and warn threat actors, in which Cyber Threat Intelligence (CTI) is a solution that satisfies the above requirements. The presence of an attacker or an actor with attacking behaviour will be identified promptly by CTI, allowing individuals and organisations to clearly understand their motives and how they attack the system (Johnson et al., 2016). From there, cyber-attacks will be proactively stopped or eliminated by CTI before they cause serious impact or damage (Alaeifar et al., 2014). Several studies have shown that investing in cyber threat intelligence is ineffective in both financial and operational aspects (Alaeifar et al., 2024). To improve the security capabilities and defence strategies of the system, understanding and applying CTI in organisations is necessary to minimise the risks from cyber threats.

This report aims to present the fundamental concepts, frameworks, and methods of Cyber Threat Intelligence. It will especially focus on its role in protecting national security in Australia. Additionally, the report will identify current challenges in CTI implementation and propose potential improvements to enhance its effectiveness in a rapidly evolving cyber threat environment.

2. Literature Review

2.1 Geopolitics

El Zant, Jaffrès-Runser & Shepelyansky (2018) describes geopolitics as the "study of the influence of a country's geographic space on its political power and its relationships with other countries". However, in an article published by Mamadouh & Dijkink (2006), they describe geopolitics as a "contentious concept", acknowledging the various definitions of the word within different realms of academic disciplines. In geographical terms, geopolitics describes the approaches to "foreign policy practices and representations", while in foreign policy studies, geopolitics describes a more conservative and realist view of international relations (Mamadouh & Dijkink, 2006).

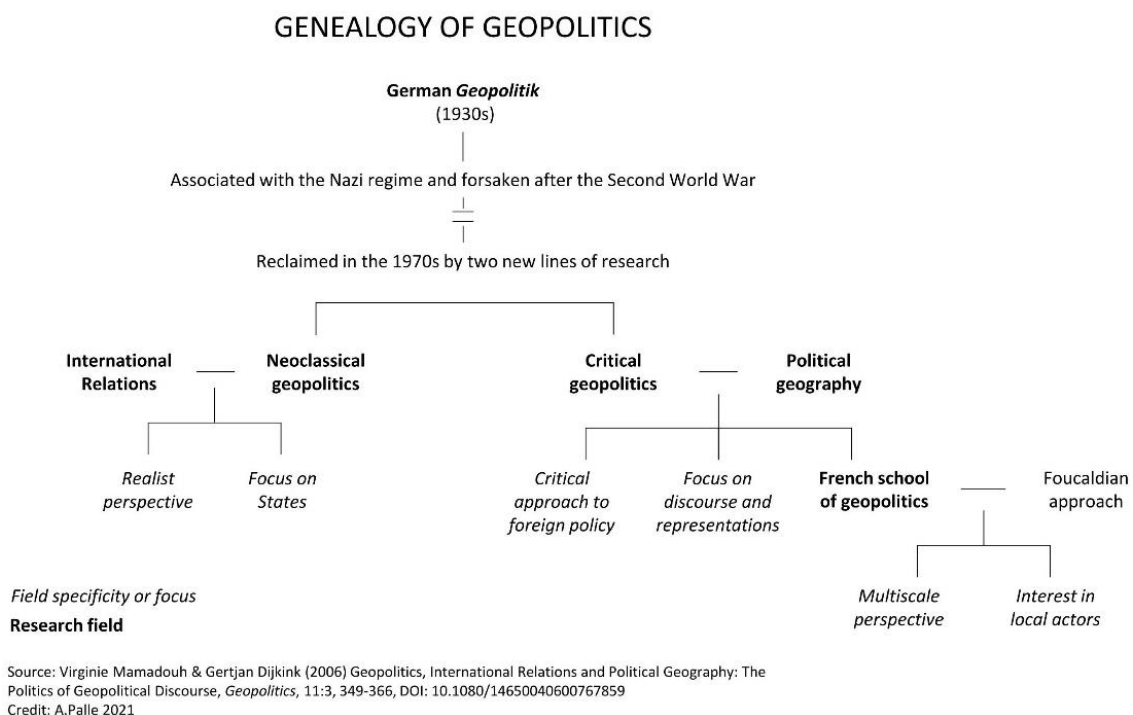


Figure 1: Genealogy of Geopolitics Field by specificity or focus and research field structure.
Source: Mamadouh & Dijkink, 2006

Political geographers developed a new approach to geopolitics in the 1990s with the goal to disclose geographical assumptions in geopolitical discourses, with particular focus on the politics and power relations behind the individuals that concern themselves in state affairs (Mamadouh & Dijkink, 2006). Based on the concepts discussed above, geopolitics can be defined as the interaction between multiple complex concepts such as geography, foreign policy, international relations, as well as politics.

In 2025, the state of international geopolitics is as complex as its definition. One of the key variables that Mucha (2025) considers important for geopolitical stability is the global power competition between China and the US. This year alone, Beijing has increasingly applied military pressure on Taiwan, as well as declaring a preferential partnership with Russia, the US's main strategic rival (Chivvis et al, 2024). In retaliation, Washington has responded with a series of tariffs and foreign policies targeted to denying China access to US technologies, as well as a diplomatic campaign to strengthen its alliances with its Asian partners (Chivvis et al,

2024). The growing instability in Asia as the escalating rivalry between the two superpowers Beijing and Washington underscores the role of the balance of power, which will be a decisive factor in shaping the international geopolitical trajectory in the coming years.

The conflict and political tensions between Russia and Ukraine have also partly affected the geopolitical situation for countries in the region and internationally in general. According to Mucha (2025), this is considered one of the current concerns in Europe. It can be said that the Russian-Ukraine conflict has changed some of the ways in which European security is operating (International Crisis Group, 2025). Until now, Europe has been almost dependent on gas fuels from Russia, about 40% of this total production depends on Russia. When the conflict occurred, Europe had to change its climate and energy policies to reduce its dependence on Russia, thereby reducing the total energy supply to almost zero (Moore, 2024). This has in turn accelerated Europe's energy transition plans forward by a decade (Moore, 2024), demonstrating the effect of geopolitical crises on policies and regulations. Beyond Europe and looking onshore, the conflict impacted Australian petrol prices, bringing the price of petrol to over AUD\$2 a litre for unleaded fuel (Rumbens, 2022). A study conducted by Izzeldin et al. (2023) also notes the impact of the war on the global financial markets, focusing specifically on the week of 24 February 2024. Their study found an instantaneous negative reaction of the invasion reflected in global markets, with volatility across a variety of commodities skyrocketing a few days after the invasion happened.

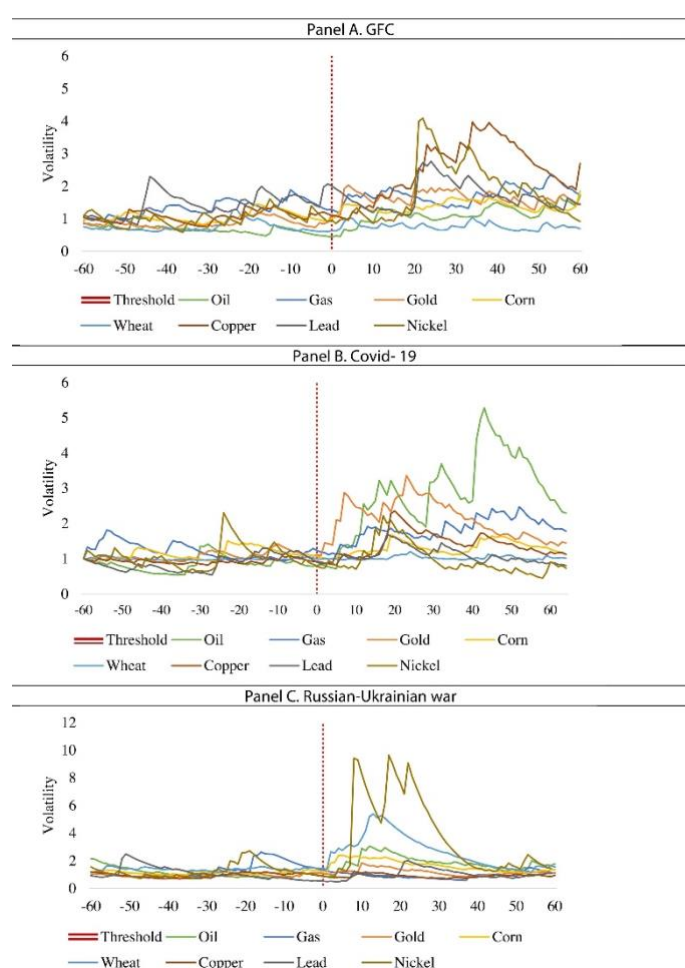


Figure 2: Conditional volatility response of commodities across 3 different crises.

Source: Izzeldin et al., 2023

In another study conducted by Deng et al. in 2022, they found that almost all countries experienced negative equity returns across the different industries in the build-up to the invasion, and companies operating in Russia, France, Germany and Italy suffered the most compared to companies in Canada and the US.

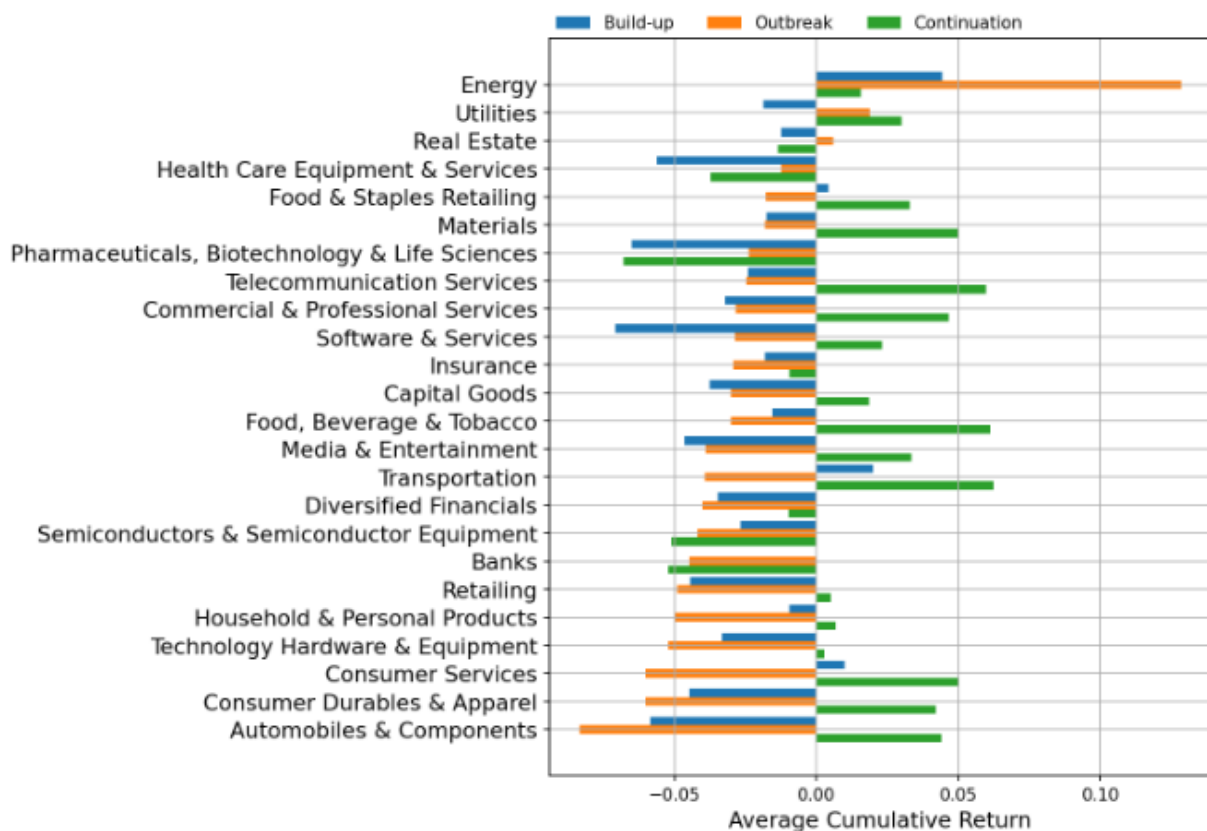


Figure 3: Stock market reactions across industries.

Source: Deng et al., 2023

In summary, geopolitics remains a highly controversial and complex topic that has a profound impact on global stability. The effects of a global competition for power, geographical land, and resources are widespread, as demonstrated by the ongoing conflicts between Russia and Ukraine and its impact on not just its neighbouring countries, but across international waters to Australia and the Americas. Its impact is not just on commodities like oil and gas, but also on financial assets and necessities like food and water, and ultimately on intangible assets such as international relations and strategic relations.

2.2 Introducing critical infrastructure

For any country, infrastructure such as energy, water, health, transportation and especially information and communication technology plays an important role in the development of the country. It is an essential part to decide the development of social security and bring a good life to the people. However, for those reasons, infrastructure is also one of the targets that attackers aim at today and is increasingly vulnerable to potential threats, potential threats from political motives as well as sophisticated cyber security threats (Alcaraz & Zeadally, 2015). Supervisory Control and Data Acquisition (SCADA) is one of the industrial control systems that plays an important role in managing essential infrastructures but has been found to contain potential risks, threats to cyber security operations including software, hardware and human factors. These vulnerabilities are particularly dangerous because they are likely to be targeted by attackers, which will quickly spread to other critical areas, and if not contained in time, the scale and severity of the disruption will be even more dangerous.

In addition, the use and integration of outdated information systems or traditional infrastructure management systems is also one of the factors contributing to the increase in cyber-attacks. The increasing number of cyber-attacks targeting critical infrastructures also poses significant challenges in the protection work because it requires consistency in operation as well as a smooth and synchronized coordination between departments in management. Fault tolerance, situational awareness, forensic investigation readiness and trust management are areas that need to be supplemented and developed to create a solid foundation against modern threats, it requires a detailed plan, a suitable strategy over a long period of time with geopolitical implications instead of individual events.

One of the important roles of Cyber Threat Intelligence (CTI) is to enhance the proactive protection of critical infrastructure systems where business continuity is a top priority for many businesses or countries. CTI can detect and analyse the attackers' targets by systematically collecting relevant information from various sources as well as ensuring the reliability of the information. From there, CTI can support the implementation of comprehensive and proactive security strategies from the early stages before an attack develops. A study has shown that realistic simulations are a useful solution to enhance incident response, which CTI can provide (Alcaraz & Zeadally, 2015). Above all, with the rapid change of security policies today, the support of a tool as CTI is extremely important and indispensable.

Because of the technological differences between infrastructures, cyber threats are becoming increasingly sophisticated and complex, and CTI today still faces difficulties in integrating. Over-reliance on tightly linked organizations has left individuals and private organizations with a lack of capacity to respond and manage their own risks. Joint ventures and international partners as well as private operators also face major hurdles due to a lack of trust in information sharing, which requires close cooperation with the host country. At the same time, the advent of AI and machine learning has opened a new era in the ability to analyse behaviour, detect abnormalities and predict, thereby providing earlier and more reliable warnings that CTI can do better than before.

2.3 Cyber Threat Intelligence (CTI) Sharing

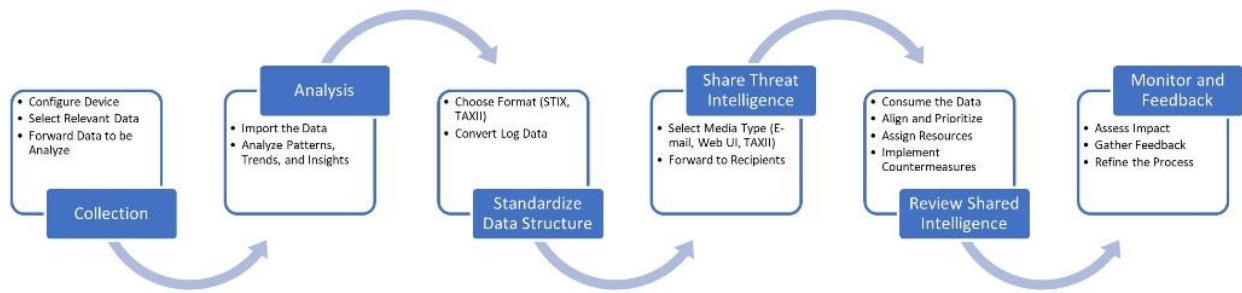


Figure 4: CTI Sharing Process.

Source: Thomas, 2023

Cyber Threat Intelligence (CTI) sharing means the sharing important threat information procedure among businesses to enhance security and support proactive protection against emerging attacks (Greyhat Infosec, 2024). Figure 4 shows the CTI sharing process initiates with collecting data from numerous sources, analysing it trends and patterns, modifying the data into a standard format for consistency, sharing the standardised data through different platforms, utilising it to guide decision-making and defences in security systems, and regularly monitoring and reviewing it to enhance its quality and applicability (Thomas, 2023).

Cyber threat intelligence helps safeguard government networks, critical infrastructure and services through timely and integrated information sharing across sectors and organisations, which enable authorities to identify threats, respond to threats effectively, and minimise the consequences from severe cyberattacks that could affect public trust and economic stability. However, Alaeifar et al. (2024) noted that organisations have encountered challenges while sharing CTI, such as legal and regulatory requirements, interoperability, and data reliability. The current environment is lack of adequate exploration, which restricts understandings of key requirements and the difficulties in the sharing process (Alaeifar et al., 2024).

Table 1: Challenges of CTI Sharing.

Challenges	Explanation
Trust	Arises from finding a balance between technology assurance, interpersonal trust, and Zero Trust principles while also considering new factors that affect sharing willingness, such as actionability, reciprocity, and fear of retaliation (Abraham et al., 2025).
Legal and Regulatory Requirements	Include diverse national laws, obligatory reporting requirements, and privacy laws such as General Data Protection Regulation (GDPR) which cause cross-border exchanges difficult and create obstacles that risk timely, coordinated cyber defence (Fang et al., 2025).
Interoperability	Caused by technological standards, strategies, policies, and legal restrictions causing it difficult for information to be used and exchanged between systems and organisations (Alaeifar et al., 2024).
Data Reliability	Challenged by the demand for timely, traceable, and integrity assured information to provide current, verifiable, and accurate data while maintaining standards, regulations and data lifecycle controls (Alaeifar et al., 2024).

These CTI sharing challenges in Table 1 have a significant impact on a nation's ability to identify, detect and respond to cyberthreats effectively. Legal and interoperability challenges restrict coordinated action across sectors and countries, whilst trust and data reliability can delay or mislead intelligence. Consequently, these restrictions could threaten national security by making critical infrastructure and services increasingly vulnerable to cyberattacks.

3. CTI concepts, types, methodologies, common frameworks

3.1 Concepts

Cyber threat intelligence (CTI) is threat data that has been collected and examined to provide cybersecurity professionals with a comprehensive picture of the possible cyberthreats to their businesses and how to prevent them (Kosinski, 2025). Organisations can take preventative measures by evaluating threat information to detect patterns, indicators, and tactics that threat actors frequently utilise. By reducing the time between threat detection and response, this changes from a reactive to a proactive security strategy helps to minimise the potential harm caused by cyberattacks (Trivedi et al., 2024).

3.2 Types

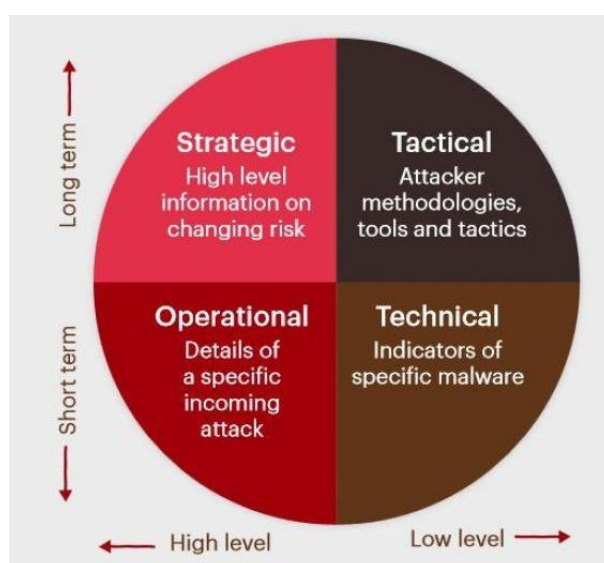


Figure 5: Types of Cyber Threat Intelligence (CTI).

Source: Karthik, 2021

Figure 5 indicates four types of cyber threat intelligence (CTI), which include strategic intelligence, tactical intelligence, operational intelligence and technical intelligence. (Trivedi et al., 2024) The long-term representatives of CTI are strategic intelligence and tactical intelligence. Strategic intelligence emphasises on the wider impacts for high-level decision-makers by analysing trends and new threats to show the possible outcomes of a cyberattack with example including policy documents and trade journals (Karthik, 2021). Tactical intelligence provides technical teams detailed insight of attackers' tactics, techniques and procedures (TTPs) and indicators of compromise (IOCs), such as IP addresses, URLs, and logs to identify and avoid breaches, and is primarily used by security personnel that are responsible for network defence (Karthik, 2021).

Whilst the short-term representations of cyber threat intelligence (CTI) are technical intelligence and operational intelligence. Technical intelligence assists in identifying social engineering attacks by focusing indications of cyberthreats, such as phishing emails and fake URLs, though its value is short life due to attackers' frequent adoption of TTPs (Karthik, 2021). Operational intelligence, which occasionally requires hacker community infiltration, offers insight into the character, purpose, timing and complexity of cyberattacks, making it a crucial but time-consuming component of a thorough threat investigation (Karthik, 2021).

3.3 CTI Sources

The Australian government operates one of the world's most comprehensive and sophisticated CTI ecosystems through a multi-layered approach that integrates domestic collection capabilities, international partnerships, commercial sources, and advance analytical platforms.

The Australian Signals Directorate (ASD) serves as the backbone of Australia's CTI capability through its comprehensive signals intelligence (SIGINT) collection infrastructure. ASD collects and analyses electronic signals from communication systems, radio frequencies and electronic data transmissions through multiple specialised disciplines (Australian Signals Directorate, n.d.). The organisation employs communications intelligence (COMINT) to gather information from foreign communications intercepted by other than the intended recipients, electronic intelligence (ELINT) which focuses on non-communication signals such as radar or energy emitted by missiles and aircraft, and foreign instrumentation signals intelligence (FISINT) derived from electronic communications like telemetry from remote system (Australian Signals Directorate, n.d.). ASD's SIGINT operations follow a structured process involving tasking (allocating resources to collect signals of intelligence value), collection (intercepting signals carrying messages or information), processing (transforming collected signals into recognisable formats), exploitation (evaluating intelligence value through analysis, decryption or translation), and dissemination (providing final SIGINT products to assessment agencies, government, and partner organisations) (Australian Signals Directorate, n.d.). The organisation has responded to over 1,100 cyber security incidents annually, with intelligence insights derived from extensive monitoring capabilities and partnerships with the National Intelligence Community (Australian Signals Directorate, 2024). ASD's technical authority position enables it to monitor cyber security threats 24 hours a day, seven days a week from a range of a local and global sources, including government, business, Computer Emergency Response Teams (CERTs), and bi-directional threat intelligence partners (Microsoft News Center, 2024).

Australian Security Intelligence Organisation (ASIO) contributes significantly to the government's CTI ecosystem through its domestic intelligence collection and investigation capabilities focusing on threats from malicious state-sponsored cyber activity. ASIO's cyber program specifically investigates and assesses threats to Australia, providing intelligence collection, investigations and intelligence-led outreach to businesses and government partners. The organisation possesses a wide range of surveillance powers to collect human and signals intelligence, including computer access warrants and surveillance device warrants where the Attorney-General determines such access will substantially assist intelligence collection relating to ASIO's functions (Merchant & Wilson, 2024). ASIO's intelligence collection, investigation and assessment efforts enable the organisation to identify threats Australia faces and provide impactful advice to inform government policy and responses to both religiously motivated and ideologically motivated violent extremism (Australian Security Intelligence Organisation, n.d.). ASIO's contribution to Australia's CTI capabilities includes providing

strategic intelligence on nation-state actors and their cyber capabilities, particularly focusing on espionage, foreign interference, and sabotage threats.

The Office of National Intelligence (ONI) operates Australia's central open-source intelligence (OSINT) capability through its Open-Source Portal, which provides registered Australian Government officials with access to current and historical open-source reporting. ONI's open-source analysts systematically collect, interpret and disseminate publicly available information relating to matters of political, strategic or economic significance to Australia to support government intelligence priorities and the work of the National Intelligence Community (NIC) (Office of National Intelligence, n.d.). ONI produces intelligence assessments on matters of political, strategic or economic significance to Australia, drawing on all sources of information available to the Australian Government including material covertly collected by other intelligence agencies, diplomatic reporting, open-source intelligence and insights gathered from world-leading experts in business, academia and think-tanks. The organisation's open-source intelligence function represents a critical component of Australia's comprehensive intelligence collection strategy, providing foundational intelligence that complements classified collection efforts. The effectiveness of ONI's OSINT capabilities is enhanced through the National Intelligence Discovery Grants (NIDG) program, which supports research addressing intelligence challenges including covert collection challenges, cyber security and protective security challenges, data-driven analytical challenges, and situational awareness assessment capabilities (Australian Research Council, n.d.).

To quickly and timely capture information reported related to cyber-attacks, governments and industries need an official communication channel, which is what the Cyber Threat Intelligence Sharing (CTIS) platform was built and put into operation (Microsoft News Center, 2024). The platform averages 129,000 attack indicators per month and has seen a sevenfold increase in shared intelligence and participation to over 50,216 as of July 2023 (Microsoft News Center, 2024). Launched in 2022, the system has already seen significant growth, highlighting the urgent need for an official communication and sharing channel between government and industry (Microsoft News Center, 2024).

According to the Australian Signals Directorate (2024), to bring efficiency to stakeholders in communication and thereby respond more quickly to threats, all useful information is aggregated from multiple sources and gathered at the central CTIS point, a tool that the government can easily automatically send this necessary information to stakeholders (Australian Signals Directorate, 2024). The platform has also been connected to major security tools such as Microsoft Sentinel and Splunk Enterprise Security to better support the information sharing process. According to the plan, from July 2025, all federal government agencies will be required to apply CTIS in security operations (Hollingworth, 2025).

Thanks to the automatic sharing form, the platform has now attracted more than 400 partners to participate and disseminate more than 1,372,400 attack indicators per year, an increase of 66% compared to before (Australian Signals Directorate, 2024). With this ability, the government can quickly provide information such as malware samples, signs of intrusion or information about attack groups, helping organizations make more effective defence and incident handling decisions.

To enhance cyber defences and understand the threat landscape in real time, the Australian government has implemented a system that monitors and filters access to high-risk domains. This system operates automatically, continuously checking traffic from suspicious websites and email servers, giving Australian Signals Directorate a full view of suspicious activity in the national cyberspace (Hendry, 2021). Over 10 billion queries have been analysed and over one million malicious connections have been blocked by this mechanism (Hastie, 2021).

The data collected by the system helps to clearly explain on how cyber-attacks occur and how attacker groups operate. Each year, the system blocks approximately 82 million visits to malicious domains, a 21% increase from the previous year (Australian Signals Directorate, 2024). It can be seen that this is not only an immediate protection measure but also a way for the government to combine active defence with intelligence gathering to improve awareness and response to cyber security threats.

Through its participation in the Five Eyes intelligence alliance, Australia has access to global cyber intelligence from partner agencies including the US National Security Agency (NSA), the UK Government Communications Headquarters (GCHQ), the Canadian Communications Security Establishment (CSE) and the New Zealand Government Communications Security Bureau (GCSB) (Australian Signals Directorate, n.d.). This collaborative platform, based on the UKUSA agreement, allows member states to share signals intelligence data and conduct joint analysis of cyber threats. As a result, the Australian Signals Directorate benefits from technology, expertise and strategic analysis capabilities that Australia would be unlikely to develop on its own in the short term (Australian Signals Directorate, n.d.).

Five Eyes cooperation has practical benefits in many areas of intelligence, including counterterrorism, cyber response, and modern regional conflict (Office of National Intelligence, n.d.). A specific example occurred in February 2024, when the countries of the bloc coordinated the release of a joint warning about PRC state-backed cyber-attack groups that were assessed to be attempting to silently penetrate information and communications technology systems in preparation for large-scale disruption operations (Australian Signals Directorate, 2024). Through this sharing mechanism, Australia not only receives intelligence on global threats but also participates directly in collective defence efforts, contributing analysis and developing joint warning reports with international partners in the Five Eyes alliance.

The Australian Criminal Intelligence Commission (ACIC) uses the special partnership with law enforcement, intelligence, and national security agencies, both domestically and internationally, to collect and share data relating to cybercrime (Australian Cyber Security Centre, 2021). Serious crimes in Australia include organised crime, drug trafficking, money laundering and especially cybercrime, which are closely monitored and investigated by the Australian Criminal Intelligence Commission (ACIC). The ACIC makes it easier to stop criminal activities thanks to the collection of information, tracking and analysis of intelligence data using CTI, then all the analysis results are shared and passed to other police agencies for coordination in investigation and prosecution.

3.4 Common Frameworks

To protect national interests and critical infrastructure, the Australian defence sector deploys CTI frameworks designed to be integrated and consistent. These frameworks, developed by the Australian Signals Directorate and the Department of Home Affairs, are continually adapted and updated to reflect rapidly evolving cyber threats.

Developed by the Australian Signals Directorate, the Australian Government Information Security Manual (ISM) provides risk-based guidance to help agencies protect data, IT systems and operational infrastructure against cyber threats. It is considered the primary source of reference on cyber security for all government agencies, including defence agencies, helping to establish a common standard to enhance defences against increasingly sophisticated cyber threats. The defence organization can accurately determine the level of threat and then apply appropriate countermeasures by regularly updating the content of the ISM to reflect new

adversary attack techniques and modes of operation. Special publications such as NIST 800-37 Version 2 are also one of the risk management frameworks that ISM is based on. This framework addresses six implementation steps including system identification, control selection, control implementation, effectiveness assessment, operational approval, and continuous system monitoring (Australian Signals Directorate, 2025).

To be effective in preventing common attacks in the current context, instead of focusing on generic solutions that lack detail on key elements, a detailed set of eight key techniques will bring more value. The core set of measures for cyber security, the Eight Essential Elements identified by the Australian Signals Bureau (ASB), includes strategies related to managing permissions for a limited number of allowed applications. In addition, it also covers operating system update and software patching processes. Administrative permission management techniques are also outlined to prevent exploitation from multiple administrative sources. Strengthening protection and prevention measures such as regularly backing up data to avoid losing important data when attacked, using user configuration tools to limit privileges, using multi-platform security measures to build a solid defence against attacks, especially blocking dangerous macros in software such as Microsoft Office.

By using CTI to understand and analyse the behaviours, attack motives and techniques used by adversaries, the Eight Essential Elements Framework is identified and helps the organization of a specific set of guidelines on how to deploy protection activities not only in a short time but also in a long and practical roadmap. This is considered a maturity process consisting of 3 levels from 0 to 3 of an active defence system where CTI has used adversary information to exploit and discover. As a result, organizations can easily control cybersecurity at the appropriate level of adaptation, application, and development for each individual system and each individual organization against sophisticated threats. The Resilient Digital Infrastructure (RDI) framework represents the Australian government's strategic approach to modernising digital infrastructure security through the adoption of contemporary security practices. The five core pillars of secure infrastructure elements, open information exchange, flexible and modular systems, adaptive threat response and scalability are addressed by the RDI framework in supporting the adoption of modern cyber resilience approaches. Aiming to protect Australia's critical digital infrastructure, the RDI framework, which is managed by the Department of Home Affairs and ASD, the concepts such as Zero Trust, Secure-by-Design and Secure-by-Default has proven to be important and necessary to protect these objectives.

By continuously verifying and not allowing any system, device, or user to be automatically trusted, Zero Trust has proven to be the right approach. For places where sensitive data and critical system infrastructure are always the target of attacks, Zero Trust shows the importance of always requiring continuous verification of all access requests regardless of their origin, especially in Defence and Security environments. If Secure-by-Design is when all stages of software design, construction, installation and maintenance pay attention to security from the beginning, Secure-by-Default focuses on installing software, all the safest security settings are automatically enabled, without the user having to adjust anything further.

The MITRE ATT&CK framework is considered a tool to assess the cybersecurity posture and help identify existing and existing weaknesses in defence systems. In Australia, the trust and integration of the framework by government agencies as well as the Australian Cyber Security Centre (ACSC) demonstrates the effectiveness and reliability of the framework. It can be said that this framework is a standard that organizations are aiming to approach because it helps organizations better understand how cybercriminals operate, thereby improving their defence capabilities. The MITRE ATT&CK framework is simply understood as attack tactics, techniques and procedures, which are drawn from reality and compiled and aggregated into a

global database. The accessibility of this framework is not limited to government agencies but is an open database for any organization, business and cybersecurity team to use. They can easily access and apply this framework to identify attack behaviours, compare current security effectiveness and exchange threat information with other organizations. In Australia, a separate summary is based on the MITRE ATT&CK framework with data that includes tactics and techniques targeting Australian networks. This makes it easier for Australian organisations to access, understand and respond to real-world risks.

4. CTI Usage

4.1 How CTI helps strengthen a nation's capabilities to identify and respond to geopolitical tensions

The increasing politicalisation and militarisation of the cyber infrastructure have brought the war zone online. With the sudden increase in persistence of cyber-attacks worldwide, nations and government bodies struggle to defend their national assets from being targeted. Nations should now drive considerable interest and resources into understanding the role of CTI in supporting national security. However, primary narratives that are largely available relating to CTI are mostly controlled by consulting private firms and vendors, focusing more on what organisations should do to implement CTI, with little to no evidence on its efficacy and whether it truly contributes to the core of CTI practices.

CTI unveils confidential or privileged information that gives organisations and governments a strategic advantage in maintaining the security of their digital landscapes. This intelligence, which can be speculative or not yet available to the public, is imperative for protecting critical assets by providing insights into emerging threats.

Preventive actions should be taken in a timely manner before potential threats arise. When vulnerabilities are exploited and attacked, if not equipped with the necessary knowledge, they are easily exploited and exploited. Security analysts use CTI as a guiding tool that provides knowledge to help them make judgments, build strategies, and improve systems. For organizations, it is an effective tool to launch initiatives, plan employee training, focus on specialized skills, enhance security, and support recovery.

For the government, CTI has the ability to collect information from many reliable sources and provide relevant intelligence to determine measures that should be deployed effectively. Because of the ability to search information from many sources and strong analytical capabilities, CTI can quickly identify objects, isolate areas, and issue warnings about a danger or suspicious behaviours. It easily identifies targets with unknown motives and reasons. With the information collected, information related to new tactics, techniques and technologies, CTI can predict and provide appropriate solutions to deploy a strong defence system that can end emerging threats.

Information leaks or unauthorized intrusions are identified and analysed by CTI and displayed to the competent authorities. The assessment of the severity of the incident or the violation of the law or the corrective measures will be considered by the authorities through the information provided by CTI. With the priority of protecting national security, the government prioritizes the use of integrated CTI to be able to respond effectively to major political fluctuations and be prepared to deal with unexpected situations.

4.2 How CTI is used in protecting Australia's national security

CTI plays an important role in safeguarding Australia's national interest, critical infrastructure, governmental institutions, and economic stability to anticipate, identify and mitigate sophisticated cyberthreats before they compromise these essential systems and services. CTI could help Australian government detect new threats and weaknesses by examining the TTPs. It also enables information exchange and cooperation across Australian agencies and stakeholders, including private-public partnerships (Dixon, 2025). Additionally, it enhances incident response and helps in attack prevention by providing information for proactive defensive strategies, including threat hunting and vulnerability patching (Dixon, 2025). Australian government is able to prevent cyberattacks, detect vulnerabilities in an early stage, and protect national security interest by incorporating CTI into national cybersecurity plan.

In fact, Australian Signals Directorate (ASD), a key member in Australia's national security community that supports the Australian Government and Australian Defence Force (ADF) (Australian Signals Directorate, n.d.), has operated a two-way sharing platform called Cyber Threat Intelligence Sharing (CTIS), which allows Australian government and business partners to receive and exchange information about malicious cyber activities (Australian Signals Directorate, 2024). CTIS can be an important strategy since the critical infrastructure and businesses could collaborate with Australian government to improve the defence techniques against cyberthreats. Therefore, they are able to identify, reduce, and response to criminal and state-sponsored cyberthreats more effectively, which could strengthen Australia's national security.

According to Australian Signals Directorate (2024), the growing participation of industry and government partners in the ASD-Microsoft initiative integrating ASD's CTIS with their Microsoft Sentinel platform shows how such inter-organisational partnerships significantly enhance the speed and volume of information sharing between and among stakeholders in both the public and commercial sectors. Such collaborations can improve the Australia's capability of responding to cyberattacks by accelerating threat detection, increasing cybersecurity awareness and strengthening coordinated defence measures.

Since its implementation, Cyber Threat Intelligence Sharing (CTIS) has increased 66% of partnerships and shared over 1,372,400 indicators of compromises (IOCs) (Australian Signals Directorate, 2024). This is critical since IOC can help Australian government and critical infrastructure to early detect, prioritise responses, improve incident response plan (IRP) and reduce future security risks (Cisco, n.d.). Hence, CTIS has significantly contributed to defending Australia's national security against future sophisticated cyberthreats.

5. Problems, challenges of implementing CTI in the current cybersecurity landscape

5.1 Global

CTI is often seen as a useful tool and a common standard for combating cyber threats, but it has also faced some significant challenges in its implementation. Scaling CTI nationwide is currently facing many difficulties, although some CTI methods are free, but overall, there are still many limitations that make the integration less effective than expected and even increase the risk.

Nations worldwide struggle with fundamental resource constraints that impede effective CTI implementation. The cybersecurity skills shortage still represents one of the biggest hurdles

globally, with studies showing that while the global cybersecurity workforce increased to 5.5 million professionals (ISC2, 2023), the workforce gap increased by nearly 19% over the same period, leaving approximately 4.8 million positions unfilled worldwide (ISC2, 2024). The complexity of modern CTI requires specialised knowledge and ongoing training that many nations lack. Government organisations worldwide compete with private sector salaries for skilled CTI analysts, and many developing countries cannot afford comprehensive educational programs or professional development pathways for CTI specialists (Wagner et al., 2019). For many nations, major skill gap issue has now caused the brain drain phenomenon, where skilled professionals migrate to higher-paying positions in more developed countries (Van Dale et al., 2023).

Effective CTI require national agencies to process enormous volumes of threat data from diverse sources, which is a huge analytical burden. Organisations struggle with filtering irrelevant or redundant data and identifying actionable intelligence in fast-paced environments, which in turn, causing delayed or missed responses to cyber incidents (Santos et al., 2025). Despite the volume of available threat data, studies reveal that only a few types of threat data objects are effectively shared, with malware signatures and URLs accounting for more than 90% of collected data, while critical information rules of attack detection comprise only 0.009% of shared indicators (Jin et al., 2024).

Effective CTI also requires seamless sharing and integration across organisations and borders, yet threat intelligence platforms remain largely non-standardised and proprietary. Isolation of information creates major intelligence silos and reduces the effectiveness of collaborative defence efforts. Frameworks like STIX and TAXII exist to help reducing integration issues, but adoption is still uneven globally, and technical barriers continue to hinder automated intelligence sharing (Santos et al., 2025).

When implementing CTI, reluctance to share information is one of the biggest challenges that parties involved in the information sharing process often face. Often, building trust in a relationship to be willing to share information takes a lot of time, meetings, negotiations, and resources. Some barriers to sharing information include government agencies often limiting the sharing of threat intelligence for fear of the information being misused or exploited through trusted relationships. This also creates conflicting opinions about whether sharing information will do more cons than pros.

Next, the complex requirements of data collection and transfer and information sharing between stakeholders are also difficult when the legal framework in different regions is different. Specifically, organizations are required to comply with the law in sharing data in a certain jurisdiction, but this legal framework does not apply to other jurisdictions. Because in many different geographical areas, they issue different laws and policies on sharing and protecting data such as GDPR in Europe or information sharing restrictions in many Asian countries. These strict regulations ensure that data sharing is done securely, but it is a big challenge and requires a lot of procedures and accuracy as well as permissions from the relevant parties. This requires a lot of time and effort for legal procedures, complex legal procedures can be a barrier to international cooperation in the field of cybersecurity. Forcing countries to be careful and consider carefully before implementing.

In the process of operating globally, some factors that affect the effectiveness of information sharing include political tensions, differences in diplomatic relations, and culture. Many countries believe that intelligence sharing has many potential reasons and so actively limit it, while others are due to diplomatic issues. Moreover, each country in the world has a different cultural orientation on cybersecurity and a different level of acceptance of information sharing, so openness in intelligence exchange relationships is extremely important. In addition,

language barriers make information sharing more complicated and difficult. With the increase of political conflicts and tensions, information sharing will become more selective in terms of trust and political relations than the need to operate cybersecurity.

5.2 Within Australia

Despite CTI not being a new concept to the Australian defence, implementing CTI solutions on a national scale can be a struggle despite all the efforts available.

Australian government agencies struggle with fragmented IT service delivery across multiple providers, leading to inconsistent performance and communication breakdowns in CTI implementations (Kinetic IT, 2024). In fact, these issues of fragmentation can bleed into the cybersecurity domain, much like what transpired with the Gateway Security Standard consultation in which disaggregated gateway and SSE architectures reduced visibility, complicated assurance processes, and made it harder to integrate CTI into day-to-day security operations (Australian Government, 2025). However, rushing and forcing integration has its own offsets, as seen during the implementation of automated CTI sharing for government agencies. Under PSPF Direction 003-2024, all Australian Government entities are required to enrol in the ASD-managed Cyber Threat Intelligence Sharing (CTIS) platform, but the consultation paper noted that there was a significant gap in monitoring capacities between entities. Many face difficulties in achieving adequate visibility, ingesting and processing large volumes of gateway logs, and integrating this data into broader event logging and retention systems. These challenges are exaggerated in disaggregated or hybrid gateway architectures, where ensuring consistent assurance and effective use of CTI becomes more complex (Australian Government, 2025).

Sharing of cyber threat intelligence through CTIS platforms has been met with controversy and resistance related to differences in cybersecurity culture and the sharing of information by organisations and companies despite government regulations. Reputational concerns or potential legal risks related to liability for enforcing regulations are what companies and organizations face when sharing incident information (Engineers Australia, 2023). Companies have said they are reluctant to share information because it amounts to “punishment for doing the right thing” from reporting regulators, which is highlighted in industry reports (Engineers Australia, 2023). Furthermore, security concerns for government agencies have also deterred organizations from engaging in intelligence sharing. To encourage voluntary sharing of information and disclosure of cyber incidents to the widest possible extent, ASD and ACSC need to ensure that their obligations regarding information security are met, as industry stakeholders have argued (Macquarie Telecom, 2023). Without such protections, there is no way to push back on the resistance towards intelligence exchange, which in turn will affect the expansion of the culture of CTI sharing.

Not only companies, agencies and businesses, but also government agencies such as the Australian Government face significant challenges in sharing CTI information while building relationships of trust with stakeholders. In the effort to improve Australia’s cyber security, information sharing is one of the time-consuming and resource-intensive issues that has become a major obstacle to improvement and growth as identified by the Australian Strategic Policy Institute (Caples, 2023). Businesses have become more cautious about sharing information about cyber incidents with the government due to reputational risks, legal consequences, and stock price declines. These concerns have prevented companies from reporting to the government despite requests from government agencies. Not taking any risks, existing government-led sharing initiatives remain heavily focused on technical threat

information rather than broader strategic alliances (Sandell, 2024). The CTIS program, while providing valuable technical indicators, has not yet achieved the comprehensive information sharing needed for effective national cyber defence. Industry feedback suggests that current programs lack sufficient detail to be actionable while sometimes using excessive jargon that makes information difficult for non-experts to understand and apply (Engineers Australia, 2023).

Government IT security teams face overwhelming volumes and complexity of threat intelligence from multiple sources. The Public Sector Network article notes that “busy and under-resourced IT security teams need to prioritise which intelligence to act upon”, and that “hundreds of indicators of compromise (IOCs) to look for” but limited capacity to investigate all threats (Anderson, 2022). It also points out that many CTI feeds are not tailored for government use, often covering mostly financial threat actors rather than focusing on state-sponsored APTs targeting government infrastructure. Adding on to that, Australia is also facing a longstanding and worsening cybersecurity skills shortage, a challenge that is particularly acute in government. ASPI notes that this gap has persisted for decades, and that below-market salaries and rigorous security clearances add further obstacles to government agencies (Caples, 2023). The shortage affects all cybersecurity across all domains, including the capacity to adopt and sustain advanced functions such as CTI, which demand intensive technical knowledge.

Implementing and maintaining CTI integration in operations requires a serious and costly investment of resources, money and time. Compared to other activities, investing in CTI requires a continuous and long-term investment, which is not considered one of the top priorities (Queensland Audit Office, 2024). Therefore, some significant limitations and difficulties are also exposed when government agencies have to consider budgets for investment in infrastructure, recruitment and training of highly skilled technical personnel. These hinder the development and integration of comprehensive CTI (Queensland Audit Office, 2024). Furthermore, the deployment of CTI platforms in many government agencies is using legacy systems that are difficult to integrate due to the gap in technology between traditional and modern (Engineers Australia, 2023). Over-reliance on legacy systems, reluctance to invest in new systems, and a lack of skills also limits the ability of agencies to deploy analytics, automation, and proactive threat intelligence tools. Cybersecurity resilience measures are identified by the Queensland Audit Office (2024) as one of the discretionary and non-core investments to the point where CTI deployment is no longer a major concern, leaving agencies vulnerable to new threats.

6. Challenges that national security faces

Cyber threat intelligence (CTI) strengthens national security by transforming collected information into actionable intelligence, which supports informed decision making by policymakers and security professionals (Ainslie et. al., 2023). CTI also helps organisations transition from a ‘reactive and undirected’ cybersecurity approach to a more ‘proactive and dynamic’ approach (Ainslie et. al., 2023).

Due to the complexity, frequency and severity of cyberattacks, it is impossible for organisations to apply standard cybersecurity procedures for a 'one-size fits all' solution, which increases the need for intelligence sharing to bolster both resilience and action when it comes to strengthening national security. WEF (2020) describes intelligence sharing as essential for generating “... the right level of situational awareness” so that an organisation can defend itself when an attack occurs. To enable this, organisations are encouraged to share a variety of information, including strategic, technical and operational intelligence across the ecosystem (WEF, 2020). The information shared by others can be used to understand the motivation behind an attack, as well as understanding what trends, techniques or tactics the attackers are using (WEF, 2020), thus allowing the organisation to be better prepared against future threats.

Research conducted by PricewaterhouseCoopers (PwC) in 2024 found that many Australian organisations listed cyber risk as their top concern, above other concerns such as geopolitical risk and inflation. PwC (2024) also found that many Australian organisations will increase their cybersecurity budgets by at least 6% by 2025. The Australian Department of Home Affairs (2024) issued a Directive in July 2024 requiring government agencies to connect to the Australian Signals Directorate (ASD) CTI sharing platform. The Directive also called on government agencies to participate in the ASD Cybersecurity Partnership program (Department of Home Affairs, 2024). This Directive helps facilitate the seamless sharing of threat intelligence between ASD and program partners, allowing organisations to better understand threat actors’ TTPs and more effective solutions to counter cyber-attacks (Anderson, 2024). This demonstrates Australia’s growing and ongoing commitment to protecting national security in the years to come.

6.1 Utilities

According to Casanovas & Nghiem (2023), cyberattacks in 2018 targeting industries including electricity, water, and gas were at an all-time high. The highest level ever recorded was in 2022, after the Russian invasion of Ukraine. Some characteristics that Bailey et al. (2020) mention that make the utilities industry particularly vulnerable to attacks include:

1. An increased number of attacks to the utilities sector by nation-state actors,
2. An increased number of attacks surfaces due to the geographical and organisational complexities, with an emphasis on the nature of many organisations’ cybersecurity teams, and
3. The interdependencies between the physical hardware and technology that allows attackers to exploit them as entry points.

The utilities sector faces increasing cyberattack risks targeting operational technology and information technology systems, making the potential for service outages, data breaches, and sabotage a growing concern amid hostile geopolitical environments (Caputzal, 2024). Some examples of technical vulnerabilities that exist within the utilities sector include:

- The use of smart grids and SCADA systems,
- The reliance on legacy equipment, and
- The increased reliance of using remote access in utility systems (Sikha & Somepalli, 2023).

In October 2024, American Water - the largest traded water and wastewater utility company in the US (Reed, 2024) - detected unauthorised activity in their internal network and systems, and as a precaution shut down their billing system and customer portal (Rosenbaum, 2024) to prevent possible further disruptions to their operations. Although no one has claimed responsibility for the attack, research found that multiple utility companies were attacked by nation-state actors from Russia, Iran, and China within the same year (Kerner, 2024). This example shows the effects of increased reliance on interconnected systems and the increase in the exposure of utility companies to external threats (Reed, 2024). Although American Water reported that no critical water operations and systems were disrupted (Reed, 2024), if unauthorised activity had not been detected, the potential consequences could have been more severe.

6.2 Education

For every individual and community, the right to education and access to education is a fundamental human right. The education sector plays an important and essential role for any country. In socio-economic development, education creates many spillover effects from improving health, reducing unemployment, increasing social cohesion to promoting technological innovation (London School of International Business, 2022). The number of cyberattacks targeting educational institutions in a single year has increased by 75% by 2025, according to Check Point's State of Cybersecurity 2025 report. Of those cyberattacks, 68% were delivered via email, spam, and junk text messages, while the remaining 32% required the use of more advanced techniques such as FakeUpdates (Check Point Team, 2025). The Parliament of Australia (2022) reported over 120 university cyberattacks in the preceding 12 months, as noted in the Australian Cyber Security Centre's Annual Cyber Threat Report 2019-20.

Educational institutions are prime targets because they store extensive sensitive personal and non-personal data, such as financial records, funding information and sensitive research data, and are particularly vulnerable to ransomware attacks, with public institutions frequently utilised taxpayer funds to cover payments (Munisteri & CISSP, 2025). Beyond immediate cybercrime, state-sponsored espionage poses a significant long-term threat, particularly to universities engaged in innovation and international research partnerships, as nation-states exploit both cyber and traditional access routes to obtain sensitive data (The National Cyber Security Centre, 2019). Differences in how universities manage and monitor international collaborations and foreign funding have created varying levels of security risks (The National Cyber Security Centre, 2019). Some current cybersecurity risks include:

- Significant disruption from the rise of generative artificial intelligence (AI) and associated security and privacy risks (Kervin, 2025),
- Increased digitisation, widespread device use, and storage of sensitive research and intellectual property (Aryon Technology Solutions, 2023), and
- Reliance on outdated network and technology infrastructure, creating security vulnerabilities exacerbated by limited funding and the continued use of legacy systems for critical functions (Aryon Technology Solutions, 2023).

A cyberattack on the medical school of the University of California in San Francisco was carried out by the Netwalker Ransomware group in June 2020 (Winder, 2020). In this attack, the hackers encrypted all the school's important data, including personal information (Winder, 2020). Initially, the hackers demanded a ransom of \$3 million in exchange for all of the data, as well as the tools to decrypt it. However, UCSF tried to contact and negotiate with the hackers on the dark web to reduce the price to \$1.14 million. Eventually, a deal between the university and the hackers was successful and the entire transaction was made in bitcoin. The UCSF data breach has brought about a negative public reaction to the ethics and behaviour of aiding and abetting a hacker group. Furthermore, the incident also raised questions about the system's ability to back up data (Winder, 2020). Instead of restoring the data internally, UCSF opted to pay the ransom, which is considered an act of complicity to cybercriminals. Weaknesses and loopholes in the management of important national assets were exposed, causing a serious decline in public trust, disrupting innovation and infringing on intellectual property rights.

6.3 Logistics

Logistics is a crucial component of supply chain management as it involves the activity of organising, carrying out, and managing the flow of products and services from point of origin to consumption (Robinson, 2023). Logistics are a major target for cybercriminals since speed, connectivity, and just-in-time delivery in logistics sector are important in a global economy (viLogics, 2025). This risk is increased by just-in-time inventory management systems as production shutdowns will cause the lack of time buffers or stockpiles, which businesses can lose millions of dollars (Mykyta, 2025). The freight and logistics sector as a critical component of the national economy contributes over 8.6% of Australia's GDP, supporting economic growth and quality of life through efficient and sustainable supply chains (Department of Infrastructure, n.d.). Its strategic and economic importance attracts state-sponsored actors, who could attempt to conduct espionage, disrupt supply chains, or weaken national resilience.

Technology integration has improved productivity in areas, including inventory monitoring and delivery management, but it has also increased exposure to cyberthreats (Iain, 2025), resulting data breaches, operational disruptions and financial losses. From 2024 to 2032, we can see that the logistics cybersecurity market is growing strongly at a CAGR of over 10% (Gujar & Kshirsagar, 2024). Of which, in 2023, the total market is valued at USD 7.25 billion. Along with the strong growth, the risks of finding profit from cyber threats are increasing. According to a report by the Cybersecurity and Infrastructure Security Agency (CISA), the sector has recorded a 50% increase in cyber incidents between 2020 and 2023, which can be attributed to the rapid digital transformation, highlighting the vulnerability of the sector to cyber-attacks (Gujar & Kshirsagar, 2024).

Key vulnerabilities include:

- Reliance on critical niche suppliers, outdated information technology and operational technology systems, which complicates efforts to enhance cyber resilience (Gogh et al., 2024),
- Greater reliance on digital technologies and increased interconnectivity, which increase vulnerability to costly cyberattacks (Gogh et al., 2024),
- Reliance on critical infrastructure, where interruptions have the potential to impact entire economies (viLogics, 2025),
- Expanded attack surfaces due to telematics, IoT devices, remote work, and warehouse automation (viLogics, 2025), and
- High financial risk, which makes ransomware attacks more profitable (viLogics, 2025).

These vulnerabilities make logistics networks attractive targets not only for cybercriminals but also for state-sponsored actors. A Russian state-sponsored cyber campaign conducted by the General Staff Main Intelligence Directorate (GRU) 85th Main Special Service Center (Military Unit 26165) (Australian Signals Directorate, 2025), also known as APT28, Fancy Bear, Forest Blizzard, or BlueDelta, has targeted Western logistics organisations involved in coordinating and distributing international aid to Ukraine since at least February 2022 (National Security Agency/Central Security Service, 2025). The campaign affected sectors including defence, transportation hubs, maritime, air traffic control, and IT services, impacting government and private organisations across air, sea, and rail transportation in NATO member nations, Ukraine, and international organisations (Australian Signals Directorate, 2025). Actors attempted to compromise logistics IT systems by deploying malware such as HEADLACE and MASEPIE, exfiltrating Office 365 user data, and maintaining access to email accounts (Australian Signals Directorate, 2025). They also targeted Internet-connected cameras in Ukraine and neighbouring NATO nations to track the movement of shipments (Australian Signals Directorate, 2025; National Security Agency/Central Security Service, 2025). These actions illustrate the significant cyberthreat to logistics networks, as disruptions to communications, IT systems and monitoring infrastructure can restrict the delivery of critical aid and harm associated support operations.

7. How CTI can address geopolitical issues

7.1 Utilities

Essential industries such as electricity and water are also gradually integrating new technologies as part of their business and management strategies. However, the dependence on technology also raises concerns about cybersecurity because this is one of the key industries that goes hand in hand with the development of social security and the economy of the country because just one disruptive incident will cause an extremely serious problem causing insecurity, violence and social stability. Therefore, essential industries such as electricity and water have become one of the top targets for emerging cyber-attacks, attacks that use sophisticated technologies and especially politically motivated attacks will be even more dangerous because of the disruption and its impact is extremely large.

For utilities such as water and electricity, CTI is used as a source of information from multiple sources to identify and analyse potential threats by looking for anomalous activity that signals an attack is about to take place. Attacker activity and motives are more easily detected if CTI is integrated into the organization's early warning and protection mechanisms. They can easily distinguish and identify comprehensively the identity and comprehensive information about the attack they are targeted for. Focusing on comprehensive analysis will help utility providers to develop timely protection solutions, making efforts to prevent an attack more effective, even politically motivated attacks before causing significant damage. (Onyeji et al., 2014).

Utilities are increasingly facing challenges to comply with stringent government regulations and legal procedures as well as international cybersecurity standards amid the ongoing political tensions globally. These attacks are seen as a tool to influence politics causing disruption and threatening social stability. Therefore, governments in many countries and international cybersecurity organisations around the world attach great importance to constantly changing and developing strict legal systems to protect national cyberspace and prevent emerging cyber threats (Saeed et al., 2023). Given the constant changes in standards as well as the complexity

and inconsistent in many countries and organisations, the importance of CTI has grown significantly. Because CTI is applied not only to detect and notify early signs of cyber security threats but also to support utility services to maintain consistent cyber security operations and easily adapt and adjust cyber security operations in accordance with changes in these standards and regulations (Saeed et al., 2023).

In addition to prevention and compliance, CTI significantly enhances incident response preparedness. When a cyber-attack occurs, traditional cybersecurity systems focus on incident resolution rather than proactively enhancing response solutions. Realistic attack scenarios including geopolitical scenarios are provided by CTI, which will help provide a more proactive approach for utility organisations in preparing effective defences (Onyeji et al., 2014). For example, an attack on the national power grid during an international conflict is a highly likely scenario that a power company might encounter that CTI simulates to help the power company prepare for. This is considered one of the valuable resources that service providers should use as a training tool for their staff on how to respond to a real-life cyber incident (Onyeji et al., 2014). The well-preparation will significantly reduce the risks, losses, and impacts of a cyberattack. Consequently, essential services are guaranteed to be uninterrupted for a long time, causing negative impacts on social stability, and are always ready for crises or political instability.

To significantly enhance critical infrastructure protection, CTI encourages organisations including utilities, government agencies, and emergency response teams to share information and collaborate (Wagner et al., 2019). A strong, connected, and collaborative cybersecurity network is a key to sharing challenges, difficulties and jointly responding to the complex, politically motivated cyber threats today.

7.2 Education

A research report has shown that to detect threats early and improve response capabilities, it is essential to collect and analyse data and information from multiple sources (Bandara et al., 2021). In the context of modern education today, more educational systems are gradually shifting to a hybrid mode, which uses and depends on many technological devices, digital platforms and many new technologies that are gradually becoming popular and developed. Therefore, CTI is considered as a tool that continuously collects data from school networks, devices used by teachers and students, online learning platforms, as well as cloud computing services applied by schools (Bandara et al., 2021). By connecting these data sources, CTI creates a more comprehensive picture of the cybersecurity situation in education. Support educational institutions in early identification of cyber security risks, cyber criminals taking advantage of political instability to conduct attacks aimed at disrupting learning, stealing personal data of students and teachers, or undermining confidence in the education system, which will contribute to undermining the sustainable development of a country.

In CTI, important data types include IP addresses with malicious behaviour, malware, and phishing emails, which will help security professionals detect and prevent cyber threats more effectively. In addition, CTI also analyses unusual behaviours to provide early warnings of new attacks, even if they have not been recorded before (Ortiz-Garcés et al., 2024). Furthermore, artificial intelligence and machine learning are integrated to enhance the ability and efficiency of behavioural analysis to quickly and accurately identify abnormal behaviours or the appearance of unwanted threats. From there, IT & cybersecurity departments and experts have full information about the sources and cause of the threat to be ready for plans to prevent and eliminate those threats early.

CTI also has many practical applications in the field of education. Automated alert systems can notify when there are signs of abnormalities, while information sharing between schools and cybersecurity organisations helps build a common defence network. This approach helps prevent attacks from spreading throughout the educational community (Johnson et al., 2016). At the same time, CTI also supports the development of training content and awareness raising for teachers, students and staff. When equipped with the knowledge and skills to recognize risks such as phishing emails or suspicious links, they will be more proactive in protecting themselves and their systems (Wagner et al., 2019).

From the discussion above, it can be seen that CTI plays a fundamental role in managing and responding to cybersecurity in education. This is increasingly significant when there are more politically motivated attacks targeting the education sector, which is inherently vulnerable but very important for social development. Through its ability to detect early, enhance analysis, promote information sharing and increase awareness, CTI helps schools and educational institutions maintain a safe and sustainable online learning environment.

7.3 Logistics

CTI provides logistics organisations with 'up-to-date contextualised intelligence' that allows them to manage its geopolitical risks (Spinks, 2024) and better identify signals for armed conflict such as increased tensions, heightened aggression and military movements (Gordon, 2025). Organisations also leverage the use of CTI to map supply chain dependencies to find points of failures (MacLellan, 2025), and use the intelligence gained to plan different shipping routes which avoids conflict zones or avoid delivering specific goods/services known to be high-value targets during high geopolitical tensions. Along the same branch, CTI helps organisations with forecasting the risk of exposure to threats within key regions, industries, or suppliers (MacLellan, 2025), which could be used to identify events that could threaten the safety of supply locations and industries, as well as establish the standing relations between rival organisations (Comte, 2022).

Within software logistics, CTI can assist with the discovery of new malware, as well as potential threat actors that are targeting specific supply chains or vendors (MacLellan, 2025). This can help with identifying the weak links in the supply chain, aid organisations with determining suspicious activity within the supply chain network before the escalation to a full-scale attack. Suspicious activities like unauthorised login attempts, unusual API behaviour or access are some typical indicators of threat actors attempting to access confidential information or cause disruptions in software logistics. The constant monitoring of vulnerabilities within the network also helps organisations better understand and address their weaknesses quickly, thus minimising the possible attack surfaces and improving their security and efficiency overall (SOCRadar, 2024).

8. Recommendations to improve existing CTI methods

Research in previous sections emphasises the advantages of Cyber Threat Intelligence (CTI) in improving threat detection and response across sectors. However, Avrahami and Zwilling (2025) indicated that its adoption remains limited due to the lack of awareness and comprehension of CTI. For instance, 2019 Ponemon Institute research found that only 26% of cybersecurity experts had awareness of CTI, and only 13% mentioned that their organisations had used CTI practices (Avrahami and Zwilling, 2025). To mitigate this, it is essential to conduct regular role-based training and education on CTI since different roles within an organisation will have distinct responsibilities. For example, SOC manager or IT manager should focus on the strategic use of CTI for decision making, SOC team should have technical training on CTI for detection and responses, legal and compliance teams should learn the CTI's legal and regulatory requirements, and all employees should be trained to recognise threats and report the suspicious activities, such as receiving malicious emails contain links. Furthermore, it is crucial to develop a cybersecurity culture in the organisations to integrate CTI practices into regular business operations, and enable employees progressively get used to threat intelligence principles. Lastly, quarterly reporting on current threats and CTI findings can strengthen the significant of CTI, ensuring that employees recognise its value in maintaining organisations' security.

The research conducted in section 5.1 indicate that threat intelligence platforms are still largely proprietary, meaning that each organisation or government will have their own processes and playbooks. The lack of a streamlined CTI process or an internationally recognised playbook for CTI is resulting in the reluctance for entities to share and trade intelligence, as they fear potential exploitation or the misuse of shared intelligence either by rivals or state actors. Building and strengthening trust is one of the effective solutions to overcome this, countries need to have negotiations, discussions and build strategic relationships between allies and adversaries. Initiatives related to CTI training and development cooperation open many opportunities in developing and strengthening internationally recognized CTI frameworks, but this needs to be through strategic foreign relations policies and unified application of common standards and techniques between countries.

Section 5.2 briefly discussed the issue that the Australian defence is facing with overlapping functions between multiple agencies, which caused integration challenges as well as communication inefficiencies between these agencies. This will, in turn, affect the entirety of the national cyber security responses. To summarise, the Australian Cyber Security Centre (ACSC) operates within the Australian Signals Directorate (ASD) yet remains organisationally separate from the Cyber and Critical Technology Intelligence Centre (CCTIC), which operates within the Office of National Intelligence (ONI), and the National Office of Cyber Security (NOCS), which operates within the Department of Home Affairs, as well as the intelligence functions that operates within the Australian Intelligence Organisation (ASIO) and Australian Federal Police (AFP). All these organisations have different scopes and jurisdictions yet share the same objective. The structural separation creates persistent coordination issues, albeit with the existence of a coordination body like the National Cyber Security Committee (NCSC).

The fragmentation in organisational management has led to a few major issues for the Australian CTI ecosystem, including overlapping responsibilities between ACSC, CCTIC, the Department of Home Affairs, ASIO and the AFP, creating inefficiencies in coordination and collaboration, duplication of analysis and communication gaps that reduced effectiveness. While ACSC serves as the nation's primary cyber security operational centre, it remains structurally separate across the broader intelligence community. The solution to address this

fragmentation would be establishing a comprehensive Cyber Fusion Centre, that combines the critical functions of the different agencies including house personnels from all cyber departments from the different agencies under unified authority structure, create unified intelligence sourced from these multi-agencies, consolidate analytics, decisions and strategies.

Due to the large volume of threat data that is gathered from various sources during the collection process, analysts can make human errors in the analysis process and provide false insights into the TTPs of the threat actors. Machine learning and AI can be employed to assist the analysis of the data. Machine learning algorithms were developed to mimic the inner workings of a human brain and can be employed in the CTI process to help analysts find patterns that human analysts would otherwise miss.

Artificial intelligence (such as generative AI and agentic AI) can be utilised in the process of filtering data and removing noise from the collected data, as well as speeding up the analysis process. But due to the frequency of hallucinations in the output of using generative AI, it is still recommended for humans to oversee and review the outputs of generative AI rather than blindly believing it.

Developing and implementing detailed policies on sharing cyber intelligence (CTI) is essential to help organisations and individuals coordinate more effectively in preventing cyber security threats. A clear policy will define the rules on who is allowed to share information, what information is allowed to be shared, and how that information is protected during the exchange process. This helps limit the risk of data loss and avoid misunderstandings or disputes between parties participating in information sharing.

In addition, to facilitate the exchange of information between parties, the use of a common data format standard such as STIX or TAXII is considered extremely necessary. Because these standards act as a tool to analyse and process information during the operation process as well as monitor and detect hidden threats quickly and accurately. In a cybersecurity environment that is always changing and full of threats like today, this is an effective solution, meeting the need for the ability to quickly respond to cyber threats.

To ensure information security, checking and verifying the source of information is extremely important and indispensable to determine whether this information is reliable and accurate. These should be mentioned in the policies and clearly require the right to privacy of personal data. Regulations on protecting sensitive data should be strictly followed to ensure the rights of all parties as well as to prevent violations of the law.

Another recommendation is to build communities and information sharing groups between organisations, businesses and state agencies. These communities help members easily exchange experiences and promptly warn about new attacks or malware, thereby enhancing the defence capabilities of the entire system. This also helps create trust and long-term cooperation between parties.

Finally, information sharing policies between parties should be monitored, reviewed and especially updated regularly so that it can easily adapt to the constant changes in the network environment as well as new technologies. Training employees to understand and implement these regulations is also important for the policy to be most effective. This will make the information sharing process more efficient, more secure and help organizations improve their defences against increasingly sophisticated cyber security threats.

9. Conclusion

In the context of escalating political conflicts, especially geopolitical tensions, technology is used as a tool to carry out attacks targeting critical infrastructure and essential utility services. Taking advantage of the close connection and linkage of information systems from business organisations, attacks targeting these targets will cause significant damage as well as many obstacles to protection and defence.

Cyber threat intelligence (CTI) is considered a rival to cyber-attacks because of its ability to process and analyse information flexibly while still ensuring the reliability and accuracy of the information. It provides intelligence to organisations about the identity of the attacker, the purpose and time of the attack. The surprise of an attack will no longer be replaced by a detailed, methodical defence strategy suitable for each type of attacker and purpose. The intelligence provided by CTI is sufficient to enable organisations, agencies, especially governments, to be ready for all types of attacks, respond quickly, and minimise significant damage.

CTI is also play an important role for critical industries such as electricity, water, and transportation because these industries are vulnerable and have serious consequences for the nation's social security in the face of attacks. Old and outdated technology is used by these sectors, systems that may contain countless hidden vulnerabilities that are easily exploited. CTI meets the need for regular monitoring and identification of signs of attacks and response readiness. In addition, the ability of organisations to recover from incidents and adapt to volatile network environments is also enhanced.

Many frameworks and policies have been developed and researched in many countries, including Australia, where CTI is deployed based on policy frameworks and standards such as ISM, Essential Eight, DSPF, and MITRE ATT&CK. The standardisation of language and procedures as well as the process of sharing information are based on the above frameworks. From there, it makes it easier to exchange information between organisations, especially communication between the government and business organisations. This is to facilitate cooperation in responding together to minimise the impact of any attacks. Risk management processes and staff training, if integrated with CTI, will promote effective management capabilities.

However, implementing CTI still faces many difficulties. Human resources are lacking and lack specialized skills. Technical infrastructure is fragmented, lacking connection and synchronization. Some organisations are reluctant to share information due to legal concerns or loss of reputation. In addition, the amount of data collected is exceptionally large but the information that is incredibly useful is limited, making analysis and action slow.

To maximize CTI's effectiveness, it is necessary to focus on developing human resources, standardizing logging and data sharing processes, applying automation to handle large volumes of information, and building trust in cooperation between agencies and businesses. When CTI becomes part of national strategy and daily operations, it will help reduce damage, improve defence capabilities, and strengthen national security in the digital age. CTI should be considered as a core competency embedded in your daily work to turn data into action quickly, accurately, and securely.

10. References

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